THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If A:B = 2:3 and B:C = 4:5, then A:B:C is: 1

(a) 8:15:8 **(b)** 8:12:15 **(c)** 2:7:5 **(d)** 2:3:5

2. 18 men can do a piece of work in 10 days. 12 men will do it in (days): 1

(a) 15 **(b)** 180 **(c)** 11 **(d)** 18

3. On a map the scale is 1 cm = 15 km. Two places 135 km apart are shown on the map at a distance (in cm) of: 1

(a) 10 **(b)** 9 **(c)** 2025 **(d)** 120

4. In a survey of 60 students, 14 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 168 **(b)** 94 **(c)** 42 **(d)** 84

5. ₹ 2400 is divided among A, B and C in the ratio 1:3:4. B's share (in ₹) is: 1

(a) 900 **(b)** 800 **(c)** 300 **(d)** 1200

6. A car moving at a constant speed covers 210 km in 3 hours. In 8 hours it will cover (km): 1

(a) 570 **(b)** 1680 **(c)** 218 **(d)** 560

7. Which of the following is the greatest ratio? 1

(a) 7:12 **(b)** 3:5 **(c)** 4:7 **(d)** 5:8

8. If A:B = 3:4 and B:C = 5:6, then A:B:C is: 1

(a) 3:4:6 **(b)** 3:9:6 **(c)** 15:20:24 **(d)** 15:24:10

9. 20 men can do a piece of work in 9 days. 15 men will do it in (days): 1

(a) 10 **(b)** 180 **(c)** 12 **(d)** 15

10. On a map the scale is 1 cm = 25 km. Two places 175 km apart are shown on the map at a distance (in cm) of: 1

(a) 8 **(b)** 4375 **(c)** 7 **(d)** 150

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Which ratio is greater: 5:9 or 3:5? 2

12. x and y are inversely proportional. When $x = 6$, $y = 20$. Find y when $x = 8$. 2

13. y is directly proportional to x. If $y = 35$ when $x = 7$, find the constant of proportionality and y when $x = 12$. 2

14. Which ratio is greater: 8:15 or 7:13? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. On a map of scale 1 cm = 2 km, a rectangular park measures 7 cm by 4 cm. Find the actual length, breadth and perimeter of the park. 3

16. A family's monthly expenditure: food 30%, rent 25%, education 20%, others 25%. Find the central angle of each sector of a pie chart. 3

17. A school bus travelling at 30 km/h takes 24 minutes to reach the school. At what speed must it travel to reach in 18 minutes? 3

OR

On a map of scale 1 cm = 3 km, a rectangular park measures 6 cm by 5 cm. Find the actual length, breadth and perimeter of the park. 3

18. A family's monthly expenditure: food 35%, rent 20%, education 15%, others 30%. Find the central angle of each sector of a pie chart. 3

OR

A school bus travelling at 36 km/h takes 25 minutes to reach the school. At what speed must it travel to reach in 20 minutes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. ₹ 5250 is divided among A, B and C such that A:B = 2:3 and B:C = 4:5. Find the share of each. 5

OR

The marks of a student are: Maths 18, Science 15, English 9, Hindi 12, SST 6. Total 60. Find the central angle of every sector of a pie chart and the percentage of the total taken by Maths. 5

20. ₹ 1500 is divided among A, B and C such that A:B = 3:5 and B:C = 5:7. Find the share of each. 5

OR

The marks of a student are: Maths 12, Science 24, English 10, Hindi 8, SST 6. Total 60. Find the central angle of every sector of a pie chart and the percentage of the total taken by Maths. 5

END OF QUESTION PAPER — Total: 40 Marks